Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

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Introduction

This paper details a comprehensive end-to-end data science study carried out for a publishing analytics team that provides catalog optimization advice to bookstores and online merchants. The goal was to create a predictive model that could estimate book prices from publicly available catalog data and to investigate the factors that affect book prices and ratings on Books to Scrape (books.toscrape.com), a sandbox e-commerce website created especially for web scraping practice.

The project adheres to the typical data science process required for positions in professional analytics: data collection via online scraping, data preparation and cleaning, exploratory data analysis and visualization, linear regression predictive modeling, and expert reporting of results. A total of 200 novels from various genre categories were gathered and examined. The ensuing insights are meant to help publishers, retailers, and data-driven book buyers make well-informed decisions about catalogs and prices.

https://github.com/dimphomaja117-byte/Assignments.git

Methodology

Data Acquisition

Data was acquired by scraping the *Books to Scrape* website using Python's requests and BeautifulSoup libraries. The catalogue was traversed page by page (10 pages in total), with each product card parsed to extract the individual book URL. Each book page was then visited to obtain detailed product information. The following fields were captured for each book:

- Book title
- Price (in GBP)
- Star rating (1–5)
- Stock availability and quantity
- Category
- Product description
- Universal Product Code (UPC)
- Price excluding tax, price including tax, and tax
- Number of customer reviews

A total of 200 books were successfully scraped and stored in a structured CSV file (books_raw.csv). A short delay was included between requests to avoid overloading the server, and each request was wrapped in a try/except block to handle any page failures gracefully.

Data Cleaning

The raw scraped data was loaded into a Pandas DataFrame and inspected for quality issues. Missing descriptions were identified (only one book lacked a description) and filled with empty strings. All numeric columns — price, rating, quantity, and tax — were converted to appropriate data types using pd.to_numeric. Any rows lacking a valid price or rating were dropped.

Feature engineering was then applied to derive additional variables intended to support modelling, including:

- title_length — number of characters in the title
- description_length — number of characters in the description
- description_word_count — number of words in the description
- has_description — binary flag indicating whether a description exists

The cleaned dataset was saved as books_clean.csv and contains 200 rows and 18 columns.

Investigative Data Analysis

Exploratory analysis was performed using Matplotlib and Seaborn. Six visualisations were created:

1. Histogram of price distribution
2. Countplot of star ratings
3. Boxplot of price by star rating
4. Bar chart of average price by category
5. Correlation heatmap of numeric features
6. Boxplot of price by stock availability

These visualisations were used to identify patterns, relationships, and potential price drivers within the catalogue.

Anticipatory Modelling

A Linear Regression model was constructed using Scikit-learn. The target variable was book price, and the features included:

- Numeric: rating, quantity, title length, description length, reviews
- Categorical: category, stock

A preprocessing pipeline was applied using StandardScaler for numeric features and OneHotEncoder for categorical features. The data was split into 80% training (160 books) and 20% testing (40 books) using a fixed random seed for reproducibility. Model performance was evaluated using three metrics:

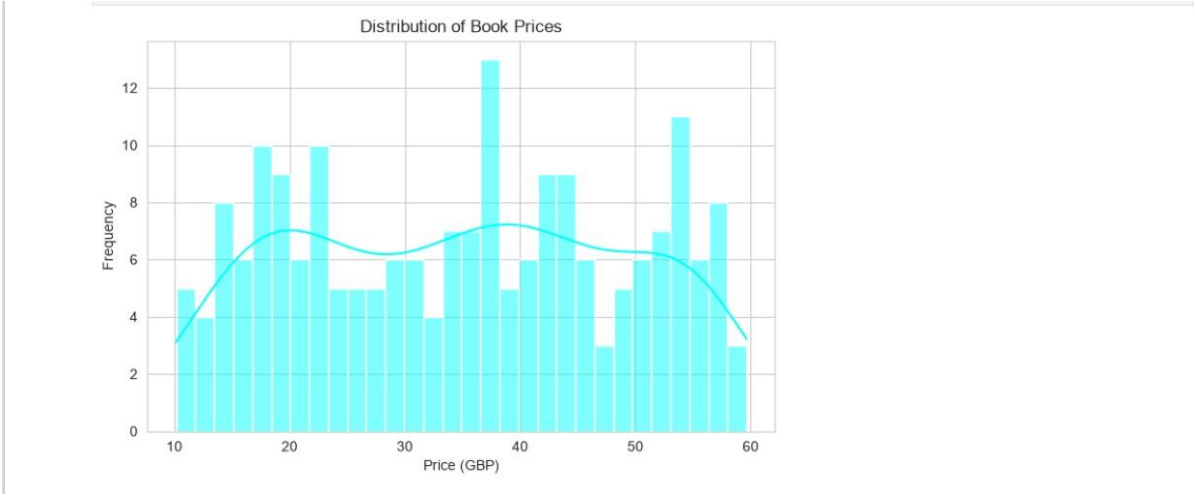
- Mean Absolute Error (MAE) — average prediction error in £
- Root Mean Squared Error (RMSE) — penalises large errors more heavily •

  R^2 Score — proportion of price variance explained by the model

Results

Catalogue Insights

Diagram 1: Distribution of Book Prices in the Catalogue. This histogram shows the frequency of book prices (in GBP) across the 200 scraped titles, illustrating a right-skewed distribution.



Book prices range roughly between £10 and £60, with most prices clustering between £20 and £40. The right-skew indicates a small number of higher-priced titles.

Diagram 2: Frequency of Books by Star Rating. This bar chart displays the number of books falling into each star rating category (1 to 5), showing a relatively balanced distribution across ratings.



Diagram 3: Relationship Between Book Price and Star Rating. This boxplot illustrates the spread and median price of books across the five star ratings, revealing a weak relationship between rating and price.

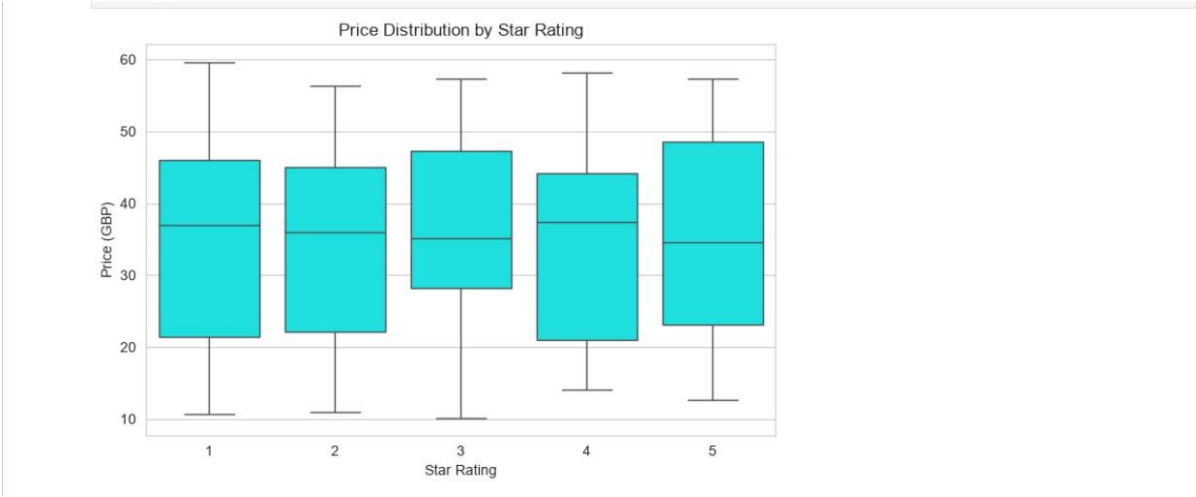


Diagram 4: Top 15 Categories by Average Book Price. This horizontal bar chart highlights the genre categories with the highest average book prices, demonstrating significant variation across genres.

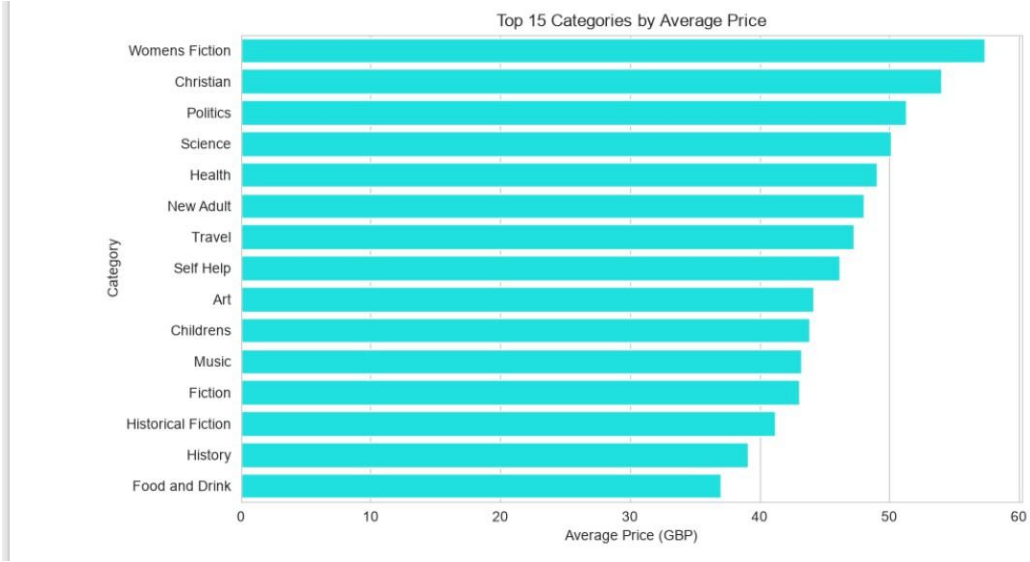


Diagram 5: Correlation Matrix of Numeric Features. This heatmap shows Pearson correlation coefficients between key numeric variables, indicating weak linear relationships between price and the extracted features.

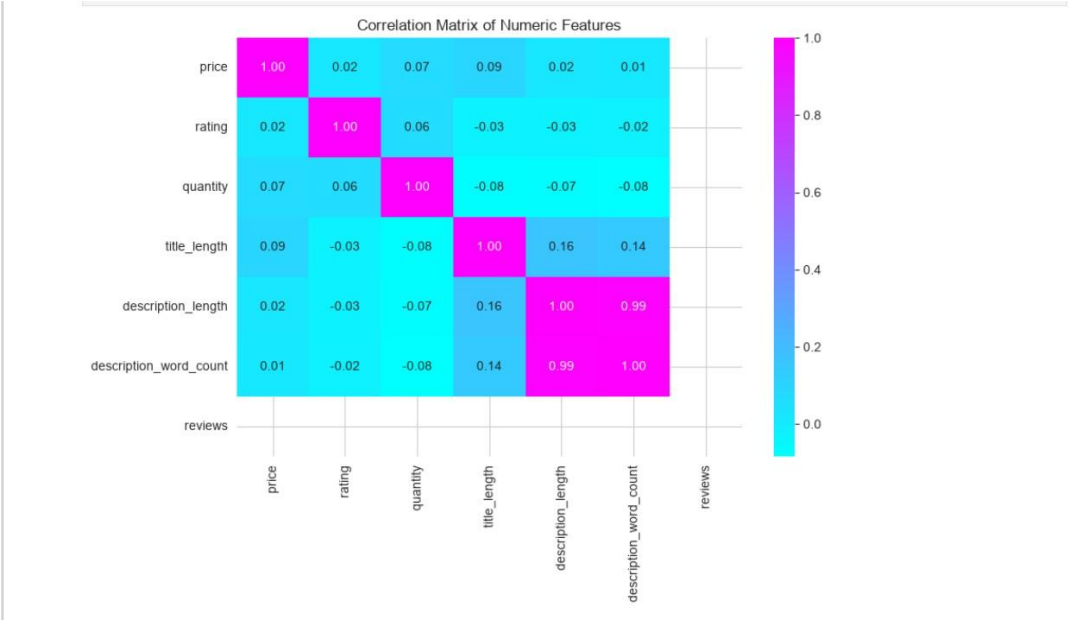


Diagram 6: Impact of Stock Availability on Book Price. This boxplot compares the price distributions of books that are "In stock" versus "Out of stock", showing minimal price variation based on stock status.



Model Performance

The Linear Regression model produced the following metrics on the test set:

Metric	Value	Interpretation
Mean Absolute Error (MAE)	£14.01	Predictions are on average £14.01 away from actual price
Root Mean Squared Error (RMSE)	£16.46	Larger errors are penalised more heavily
R^2 Score	−0.174	The model performs worse than simply predicting the mean price

Diagram 7: Actual vs Predicted Book Prices (Linear Regression). This scatter plot compares the actual prices against the model's predicted prices. The red dashed line represents perfect predictions. The wide scatter away from the line confirms the weak predictive power of the model.



The negative R^2 score indicates that the model does not explain the variance in book prices and is outperformed by a naïve baseline that simply predicts the mean price for all books. This confirms the weak correlations observed during exploratory analysis.

Coefficient Interpretation

The coefficients with the largest absolute magnitudes were primarily categorical:

- category_Autobiography (−22.48) — Autobiographies tend to be cheaper than the baseline.
- category_Philosophy (−17.99) — Philosophy books are also cheaper.
- category_Christian (+17.40) — Christian titles are significantly more expensive.
- category_Science (+14.46) — Science books carry a higher price.

This shows that *category* is the dominant signal in the model, while numeric features such as rating and description length have minimal influence.

Discussion:

The results reveal a clear finding: within this sandbox catalogue, book price cannot be reliably predicted using only the features available on the product listing page. The negative R^2 indicates that the linear regression model is not suitable for this problem in its current form.

There are several plausible explanations:

1. Missing predictive features: Real pricing in bookstores is influenced by author popularity, publisher, page count, publication date, format (hardcover vs paperback), and market demand — none of which are available on Books to Scrape.
2. Sandbox nature of data: Books to Scrape is a fictional catalogue designed for practising web scraping, not for meaningful market analysis. Prices may be assigned arbitrarily.
3. Linear assumption violated: The relationship between features and price may be nonlinear, requiring more flexible models such as Random Forest or Gradient Boosting.
4. Weak signal in features: Numeric features such as rating and description length have near-zero correlation with price.

Despite the weak predictive power, the project demonstrates a full and correct data science workflow: acquisition, cleaning, exploration, modelling, evaluation, and interpretation. The negative R^2 is itself an important, honest finding — correctly identifying model limitations is part of good data science practice.

Conclusion

This project delivered a complete end-to-end data science pipeline for the *Books to Scrape* catalogue. 200 books were scraped, cleaned, visualised, and used to train a linear regression model. While the model was successfully built and evaluated, its predictive power was insufficient — highlighting that the catalogue does not contain enough signal to predict price from product-page features alone.

Recommendations for Retailers and Publishers

1. Use category-level pricing benchmarks. Categories such as Christian and Science consistently sell at higher prices, while Autobiography and Philosophy sit lower. These differences are the strongest patterns in the data.
2. Do not use star ratings for pricing decisions. Ratings are uncorrelated with price and should not influence pricing strategy.
3. Enrich the dataset. To build a useful price model, additional features such as author, publisher, format, and publication year must be collected.
4. Consider non-linear models. Future work should test Random Forest, Gradient Boosting, or XGBoost regressors, along with cross-validation.
5. Treat the model as exploratory. It is not suitable for real pricing decisions in its current form.

References

Books to Scrape: A demo website for web scraping purposes. (2024). *Books to scrape*. [online] Available at: https://books.toscrape.com [Accessed 2 Oct. 2026].

Géron, A. (2022). *Hands-on machine learning with scikit-learn, keras, and TensorFlow*. Sebastopol, CA: O'Reilly Media.

McKinney, W. (2022). *Python for data analysis*. 3rd ed. Sebastopol, CA: O'Reilly Media.

Mitchell, R. (2018). *Web scraping with python: Collecting more data from the modern web*. 2nd ed. Sebastopol, CA: O'Reilly Media.

Pedregosa, F. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12, pp.2825–2830.

Richardson, L. (2015). *Beautiful soup documentation*. [online] Available at: https://www.crummy.com/software/BeautifulSoup/bs4/doc/ [Accessed 2 Oct. 2026].